

Natural killer cell activity in peripheral blood of highly trained and untrained persons.

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Natural killer (NK) cell activity and concentration of CD16+ cells (NK cells) and CD20+ cells (monocytes) in peripheral blood were measured in highly trained racing cyclists and in age- and sex-matched untrained controls. Median NK cell activity was 38.1% (range 20.0%-57.1%) in trained vs 30.3% (range 19.7%-43.1%) in untrained ($P = 0.008$). Median %CD16+ cells was 17% (range 7%-33%) in trained vs 11% (3%-29%) in untrained ($P = 0.007$). Indomethacin in vitro enhanced the NK cell activity in both groups. There was, however, no significant difference between the NK cell activity in trained and untrained after exposure to indomethacin in vitro. Indomethacin-enhanced NK cell activity was 45.9% (range 24.4%-67.5%) in trained and 40.0% (range 23.9%-68.5%) in untrained ($P = 0.138$). Mean %CD14+ cells was 8.3% (range 2%-15%) in trained vs 3.8% (2%-8%) in untrained (P less than 0.0001). The increased NK cell function thus demonstrated in highly trained persons might result in better resistance against infectious disease.